

Overview of Stochastic gradient descent

Subject: Stochastic gradient descent

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$$

1.

Further proposals include the momentum method or the heavy ball method, which in ML context appeared in Rumelhart, Hinton and Williams' paper on backpropagation learning and borrowed the idea from Soviet mathematician Boris Polyak's 1964 article on solving functional equations. Stochastic gradient descent with momentum remembers the update Δw at each iteration, and determines the next update as a linear combination of the gradient and the previous update:

2.

Adam (short for Adaptive Moment Estimation) is a 2014 update to the RMSProp optimizer combining it with the main feature of the Momentum method. In this optimization algorithm, running averages with exponential forgetting of both the gradients and the second moments of the gradients are used. Given parameters $w(t)$ and a loss function $L(t)$, where t indexes the current training iteration (indexed at 1), Adam's parameter update is given by:

3.

$G_{j,j} = \sum_{\tau=1}^t g_{\tau,j}^2$. This vector essentially stores a historical sum of gradient squares by dimension and is updated after every iteration. The formula for an update is now

4.

This equation is implicit since w_{new} appears on both sides of the equation. It is a stochastic form of the proximal gradient method since the update can also be written as:

5.

RMSProp has shown good adaptation of learning rate in different applications. RMSProp can be seen as a generalization of Rprop and is capable to work with mini-batches as well opposed to only full-batches.

6.

$$Q(w) = \sum_{i=1}^n Q_i(w) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (w_1 + w_2 x_i - y_i)^2$$

7.

$$[\mathbf{w}_1 \mathbf{w}_2] \leftarrow [\mathbf{w}_1 \mathbf{w}_2] - \eta \left[\frac{\partial}{\partial \mathbf{w}_1} \frac{\partial}{\partial \mathbf{w}_2} \ell(\mathbf{w}_1 + \mathbf{w}_2 \mathbf{x}_i - y_i) \right] = [\mathbf{w}_1 \mathbf{w}_2] - \eta \left[2(\mathbf{w}_1 + \mathbf{w}_2 \mathbf{x}_i - y_i) \mathbf{x}_i \right]. \quad \{\displaystyle \begin{matrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{matrix} \}$$

$$\begin{matrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{matrix} \leftarrow \begin{matrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{matrix} - \eta \left[\frac{\partial}{\partial \mathbf{w}_1} \frac{\partial}{\partial \mathbf{w}_2} \ell(\mathbf{w}_1 + \mathbf{w}_2 \mathbf{x}_i - y_i) \right] = \begin{matrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{matrix} - \eta \left[2(\mathbf{w}_1 + \mathbf{w}_2 \mathbf{x}_i - y_i) \mathbf{x}_i \right].$$

Note that in each iteration or update step, the gradient is only evaluated at a single $\mathbf{x}_i$. This is the key difference between stochastic gradient descent and batched gradient descent. In general, given a linear regression $\hat{y} = \sum_{k=1}^m \mathbf{w}_k \mathbf{x}_k$ problem, stochastic gradient descent behaves differently when $m < n$ (underparameterized) and $m \geq n$ (overparameterized). In the overparameterized case, stochastic gradient descent converges to $\arg \min_{\mathbf{w}} \|\mathbf{w}^T \mathbf{x}_k - y_k\| \forall k \in 1:n$. That is, SGD converges to the interpolation solution with minimum distance from the starting $\mathbf{w}_0$. This is true even when the learning rate remains constant. In the underparameterized case, SGD does not converge if learning rate remains constant.

8.

$$\mathbf{w}^{\text{new}} := \arg \min_{\mathbf{w}} \left\{ Q_i(\mathbf{w}) + \frac{1}{2\eta} \|\mathbf{w} - \mathbf{w}^{\text{old}}\|^2 \right\}.$$

9.

Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a "second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss

10.

$$\mathbf{v}(\mathbf{w}, t) := \gamma \mathbf{v}(\mathbf{w}, t-1) + (1-\gamma) \left(\nabla Q_i(\mathbf{w}) \right)^2$$

11.

where $\mathbb{E}$ denotes taking the expectation with respect to the random choice of indices in the stochastic gradient descent scheme. Since this approximation does not capture the random fluctuations around the mean behavior of stochastic gradient descent solutions to stochastic differential equations (SDEs) have been proposed as limiting objects. More precisely, the solution to the SDE

12.

Stochastic gradient descent (often abbreviated SGD) is an iterative method for optimizing an objective function with suitable smoothness properties (e.g. differentiable or subdifferentiable). It can be regarded as a stochastic approximation of gradient descent optimization, since it replaces the actual gradient (calculated from the entire data set) by an estimate thereof (calculated from a randomly selected subset of the data). Especially in high-dimensional optimization problems this reduces the very high computational burden, achieving faster iterations in exchange for a lower convergence rate. The basic idea behind stochastic approximation can be traced back to the Robbins–Monro algorithm of the 1950s. Today, stochastic gradient descent has become an important optimization method in machine learning.